Product Design and Manufacturing

Assignment Week-12

1. The primary purpose of reverse engineering is to assess dimensional accuracy, aiding in product improvisation and flow determination.
   a) True
   b) False

2. What is the initial step in the reverse engineering process?
   a) Creation of a CAD model
   b) Prototype development
   c) Digitization of the object/data capturing
   d) Processing of measured data

3. What is a characteristic of the scanning process using CMM-based scanner?
   a) Rapid scanning
   b) Suitable for soft materials
   c) Comparatively slow process
   d) Limited to shiny surfaces

4. Shiny surfaces and surfaces parallel to the light axis can be accurately scanned using noncontact scanner.
   a) True
   b) False

5. What are the two typical types of 3D data that the RE data processing chain can produce?
   a) Curves and 2D images
   b) Polygons and curves
   c) Points and lines
   d) 2D images and polygons

6. _____ in post-processing involves the detection and fixing of missing parts on the mesh surface by automatically filling them with a surface composed of triangles.
   a) Make manifold
   b) Fill holes
   c) Detect and repair intersection
   d) None of these

7. Coordinate Measuring Machines (CMM) provide more accurate measurement data than articulated mechanical arms.
   a) True
   b) False

8. How do most laser scanners determine the surface coordinates of an object?
   a) Through mathematical calculations
   b) Geometric triangulation
   c) Light absorption
   d) Surface scanning

9. Benchmarking is a continuous process of comparing company's:
   a) Strategy
   b) Products
   c) Process
   d) All of these

10. Which of the following deals with providing customized products to individual clients without telling them that the items that are exclusively produced for them?
   a) Cosmetic customization
   b) Transparent customization
   c) Collaborative customization
   d) Adaptive customization